

PALMER STATION, Antarctica

WMO: 890610

Lat: 64.774S

Lon: 64.053W

Elev: 8

StdP: 101.23

Time Zone: -3.00 (W03)

Period: 04-19

WBAN: 89061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-18.0	-15.0	-20.6	0.6	-17.5	-17.6	0.8	-13.8	20.8	-1.0	18.9	-1.0	4.7	240	0.959

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	5.7	3.4	5.0	2.9	4.4	2.5	3.9	5.1	3.4	4.5	2.9	4.0	6.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.9	4.7	3.9	2.3	4.5	3.4	1.8	4.3	3.1	16.9	5.3	15.9	4.7	15.1	4.1	7.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.1	12.9	-18.7	7.6	3.1	0.8	-20.9	8.2	-22.7	8.7	-24.4	9.1	-26.6	9.7		
(5)			-18.5	5.1	3.1	0.8	-20.8	5.7	-22.6	6.1	-24.4	6.6	-26.6	7.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	1.9	1.8	1.1	-1.0	-1.8	-3.5	-5.3	-6.0	-4.6	-2.4	-0.7	0.9	(6)
(7)		DBStd	3.90	1.33	1.29	1.67	2.24	2.47	2.81	3.79	4.42	4.51	3.27	2.24	1.54	(7)
(8)		HDD10.0	4248	251	228	276	328	366	407	473	496	437	386	321	282	(8)
(9)		HDD18.3	7290	509	462	534	578	624	657	731	754	687	644	571	540	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	3.2	3.9	4.5	4.8	5.0	5.0	4.8	4.8	5.3	5.0	4.5	3.4	(14)
(15)	Precipitation	PrecAvg	929	77	77	103	89	87	71	68	60	69	88	75	65	(15)
(16)		PrecMax	1247	115	130	156	146	160	120	123	138	114	146	130	115	(16)
(17)		PrecMin	672	49	45	45	54	28	37	27	20	22	51	36	27	(17)
(18)		PrecStd	140	16	20	29	25	33	24	23	25	21	22	22	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	6.3	6.5	5.1	4.6	3.9	2.9	3.1	3.5	4.3	5.2	6.2	(19)
(20)			MCWB	3.7	4.0	4.5	3.4	2.8	2.1	1.1	1.5	1.8	2.4	2.9	3.2	(20)
(21)		2%	DB	5.5	5.4	5.1	4.0	3.6	2.5	1.5	2.0	2.4	3.3	4.0	4.9	(21)
(22)			MCWB	3.1	3.3	3.3	2.7	2.1	1.2	0.4	0.7	1.0	1.8	2.1	2.3	(22)
(23)		5%	DB	4.8	4.6	4.3	3.0	2.6	1.3	0.8	1.0	1.5	2.5	3.0	4.1	(23)
(24)			MCWB	2.7	2.7	2.7	1.8	1.3	0.3	-0.1	-0.2	0.4	1.2	1.4	1.9	(24)
(25)		10%	DB	4.1	3.9	3.6	2.2	1.7	0.3	-0.1	0.1	0.7	1.7	2.2	3.4	(25)
(26)			MCWB	2.1	2.3	2.3	1.1	0.5	-0.5	-0.9	-0.9	-0.1	0.6	0.8	1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	4.8	4.6	3.7	3.2	2.2	1.6	1.8	2.1	2.9	3.6	3.9	(27)
(28)			MCDW	5.9	5.8	6.0	4.6	4.3	3.2	2.2	2.6	3.3	3.9	4.6	5.5	(28)
(29)		2%	WB	3.6	3.7	3.6	2.8	2.2	1.2	0.7	0.9	1.1	2.0	2.5	2.8	(29)
(30)			MCDW	4.9	4.9	4.7	3.7	3.2	2.3	1.4	1.8	1.9	3.1	3.5	4.2	(30)
(31)		5%	WB	3.1	3.1	3.0	2.1	1.5	0.4	0.0	0.6	1.4	1.6	2.2	(31)	
(32)			MCDW	4.4	4.2	4.0	2.9	2.4	1.2	0.7	0.8	1.3	2.3	2.7	3.6	(32)
(33)		10%	WB	2.5	2.6	2.4	1.2	0.6	-0.4	-0.9	-1.0	0.0	0.7	1.0	1.7	(33)
(34)			MCDW	3.8	3.6	3.4	2.0	1.5	0.3	-0.1	0.0	0.7	1.6	2.0	3.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.7	2.6	2.6	2.5	2.7	2.9	3.9	4.6	4.9	4.1	3.4	2.9	(35)
(36)			MCDWR	3.7	3.8	3.6	3.6	3.7	3.6	4.1	3.8	4.8	4.1	4.4	4.1	(36)
(37)			MCWBR	2.7	2.7	2.8	3.0	3.2	3.0	3.7	3.3	4.4	3.3	3.3	2.8	(37)
(38)		5% WB	MCDWR	3.4	3.4	3.5	3.6	3.6	3.6	3.9	4.5	5.1	4.0	4.0	3.6	(38)
(39)			MCWBR	2.6	2.6	2.9	3.2	3.2	3.1	3.7	4.1	4.9	3.3	3.1	2.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.287	0.275	0.248	0.360	N/A	N/A	0.242	0.264	0.284	0.286	0.290	(40)	
(41)		taud	2.389	2.409	2.417	2.311	2.885	N/A	N/A	2.165	2.276	2.274	2.326	2.356	(41)	
(42)		Ebn at Noon	975	924	839	660	361	N/A	N/A	631	826	912	972	988	(42)	
(43)		Edh at Noon	105	91	71	47	22	N/A	N/A	51	79	103	111	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.76	3.97	2.11	0.77	0.17	0.04	0.11	0.54	1.78	3.73	5.62	6.64	(44)	
(45)		RadStd	0.33	0.27	0.15	0.06	0.01	0.00	0.01	0.04	0.15	0.25	0.43	0.58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page